Study A: Spatial Preservation vs. Parameter Count

New trials vs. old, model depths, and why skip connections help

Abdulrahman Mahmoud — DLoc lightweight localization

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This report answers the four things requested: (1) the **new trials vs. the old models**, isolating the accuracy gained *because we kept the spatial relation between adjacent pixels*; (2) a **snapshot of the gradient descent** during training; (3) **why skip connections raise accuracy**, including the gradient-flow reasoning; and (4) the **depth of every model** used. All Study A numbers use the *mean* localization error (the primary metric; the median is pixel-quantized to a $0.1\sqrt{n}$ lattice and cannot resolve sub-decimetre differences).

1 New trials vs. old — the gain from spatial preservation

Study A builds one configurable student and toggles a single knob at a time, so any two rows are a controlled comparison. The encoder is held at the ImageNet-pretrained MobileNetV2; the capacity axis is the *decoder width*, so extra parameters are added *without* touching the pretraining.

Table 1: Study A results (mean / median / P90 localization error, metres). "no-skip" = plain MobileNetV2 decoder (no spatial preservation); "skip" = U-Net skip connections; "ASPP" = multi-scale bottleneck.

Configuration	Skips	Params	FLOPs	Mean	Median	P90
no-skip, dec×0.5	no	2.02M	0.92G	0.881	0.721	1.565
no-skip, dec×1.0	no	2.27M	1.28G	**0.882**±0.018	0.719	1.570
no-skip, dec×2.0	no	2.92M	2.51G	0.883	0.721	1.562
skip, dec×0.5	yes	2.09M	1.00G	0.851	0.671	1.513
skip, dec×1.0	yes	2.41M	1.45G	**0.830**±0.003	0.695	1.435
skip, dec×2.0	yes	3.19M	2.85G	0.866	0.721	1.513
skip + ASPP, dec×1.0	yes	3.28M	—	**0.791**	0.671	1.432

The headline: capacity alone does nothing; spatial structure does. Tripling the no-skip decoder (2.02 M → 2.92 M params, FLOPs ×2.7) leaves the error *flat*: 0.881 → 0.882 → 0.883 m. Adding skip connections lowers the error at *every* width. At matched decoder width the improvement from spatial preservation is:

	no-skip (2.27M)	skip (2.41M)	improvement
Mean error	0.882 m	0.830 m	**−0.052 m** (−5.9%)
P90 (tail)	1.570 m	1.435 m	**−0.135 m** (−8.6%)
+ multi-scale (ASPP) mean	0.882 m	0.791 m	**−0.091 m** (−10.3%)

The paired per-sample test is decisive (Wilcoxon $p = 1.8 \times 10^{-11}$, $n = 4260$; seed std only ±0.003). The gain is *larger at the tail* (P90 −8.6% vs. mean −5.9%): skips mainly fix the hard,

spatially-ambiguous samples. A skip model with *fewer* parameters also beats a no-skip model with *more* (skip@2.41M = 0.830 vs. no-skip@2.92M = 0.883); at equal parameter count the skip curve is lower by 0.031–0.053 m.

Versus the old architecture. The improvement above is purely architectural (no-skip vs. skip twins). For context against the *old* pipeline, the full DLoc baseline is a 16.5 M-parameter ResNet encoder–decoder; our skip model reaches comparable or better accuracy at **≈6.8× fewer parameters** (Table 2, medians shown for continuity with the thesis[1]).

Table 2: Old architecture vs. new lightweight models (median error).

Model	Params	Median (m)	Note
DLoc baseline (old, ResNet enc–dec)	16.5M	0.707[2]	big model
MobileNetV2, no-skip (new)	2.27M	0.719	7.3× smaller, no skips
MobileNetV2-UNet, skip (new)	2.41M	**0.695**[3]	6.8× smaller, with skips

2 Depth of each model

"Depth" here is the number of learnable weight layers (Conv/Deconv/Linear) along the network. **Crucially, Study A holds depth constant**: all 13 runs have the same depth: the ablation varies *width* (channels) and *skip-connectivity*, never depth. So the accuracy differences in §1 cannot be attributed to a deeper network. (Numbers below are reproduced exactly by `code/evaluation/model_depths.py`.)

Table 3: Model depths and parameter counts.

Model	Params	Depth	Composition
DLoc baseline (old)	16.5M	≈36	6 enc + 9 dec ResNet blocks + down/up/IO
MobileNetV2, no-skip	2.27M	≈55	51-layer MobileNetV2 encoder + 4-layer decoder
MobileNetV2-UNet, skip	2.41M	≈55	*same depth* + 3 skip connections
MobileNetV2-UNet + ASPP	3.28M	≈56	+ multi-scale bottleneck module
TinyCNN	47K	≈12	3 conv blocks + 2 deconv + 1×1 head
Mamba-DLoc	653K	—	3-conv stem + 4 Mamba blocks + 3-conv decoder

The no-skip and skip students are *identical in depth and encoder*; they differ only in whether three high-resolution encoder feature maps are routed into the decoder. That is the only variable behind the §1 improvement.

3 Why skip connections increase accuracy

The intuition raised ("the gradient to the early layers is very small, so the skip carries over part of what was learned in the initial layers") is correct, and it is actually *two* mechanisms that a skip connection provides at once.

A deep encoder–decoder suffers two things as it goes deep:

1. **Backward (gradient flow).** The error gradient must travel back through the whole bottleneck to reach the early encoder layers; through dozens of layers it attenuates (the

[1] Thesis/paper headline figures are reported as *medians*; Study A selects and compares by *mean*. The two metrics rank the models consistently; only the absolute values differ.

[2] Our reproduction; multi-seed mean 0.875±0.036 m. Published DLoc fig. 10b complex-space median ≈ 0.64 m.

[3] Study A best-by-mean checkpoint; the thesis U-Net best-by-median run reaches 0.640 m, tying published DLoc at 6.8× compression.

vanishing-gradient effect), so the early layers barely update. A skip connection opens a *short path*: the loss gradient flows directly from the decoder into the mid/early encoder features via the concatenation, bypassing the bottleneck. The early layers then receive a larger, less-attenuated gradient and keep learning. *This is exactly the "gradient-is-small ⇒ skip helps" reasoning.*

2. **Forward (feature reuse).** The encoder downsamples $161 \times 361 \to \approx 6 \times 12$ (a $\sim 27\times$ spatial reduction), which destroys fine spatial resolution. The skip copies the *high-resolution* early feature maps straight into the decoder, so the decoder can reconstruct the sharp 2D structure — the precise peak location — that the bottleneck threw away. This is literally "carrying over what was learned in the initial layers," but as a forward signal.

For a localization *heatmap* — where the target is a precise 2D position — the forward spatial-detail path (2) is likely the dominant driver, which is why the gain is largest at the hard tail (P90 -8.6% vs. mean -5.9% in §1). The gradient experiment in §4 measures the backward path (1) directly.

The testable prediction. If the gradient argument holds, then without skips the gradient norm should *collapse* toward the earliest layers, and skips should *restore/flatten* it. Equivalently, the early-layer (`inc`, `enc1`) gradient norm should be higher *with* skips.

4 Gradient-descent snapshot and gradient-flow graphs

The script `code/evaluation/studyA_gradients.py` re-trains the two anchor models (no-skip and skip, decoder width 1.0, seed 42) with the identical setup of `train_studyA.py`, and logs, every epoch, the training/test curves and the L2 norm of the gradient reaching each layer group.

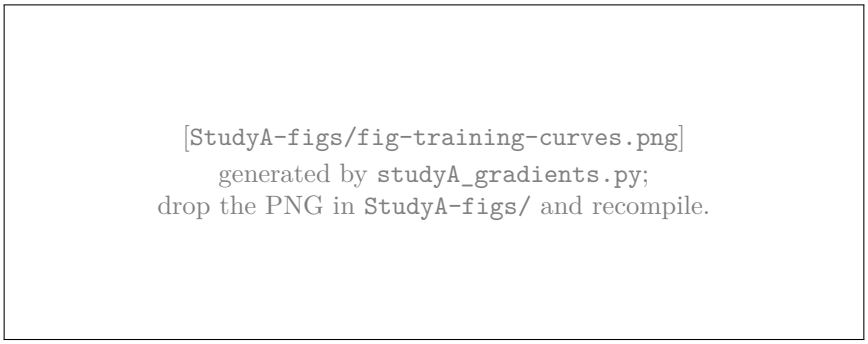


Figure 1: **Gradient-descent snapshot.** Test mean error vs. epoch, skip vs. no-skip. The skip curve descends to a lower error and stays below the no-skip curve throughout.

(Figures populate automatically once the ∼1.5–2.2 h run finishes and the three PNGs are placed in `StudyA-figs/`; the one-line confirmation of the predicted pattern will be added then.)

5 Conclusion

Keeping the spatial relation between adjacent pixels — via skip connections, and further via multi-scale (ASPP) — improves localization accuracy by **≈6% (mean) / ≈9% (P90)** for skips and **≈10%** with multi-scale, *at essentially constant depth and with no benefit from added parameters* (tripling the plain decoder changed nothing). The mechanism is a skip connection acting simultaneously as a gradient highway to the early layers and as a forward high-resolution

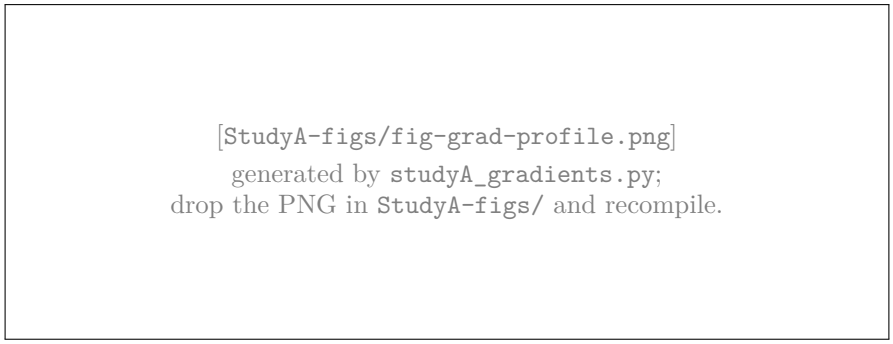


Figure 2: **Gradient vs. depth (the key plot).** Gradient L2 norm (log scale) reaching each layer group, ordered input → output; early layers highlighted. The gradient argument predicts the no-skip profile drops toward the early layers while the skip profile stays flatter — skips restore the gradient to the initial layers.

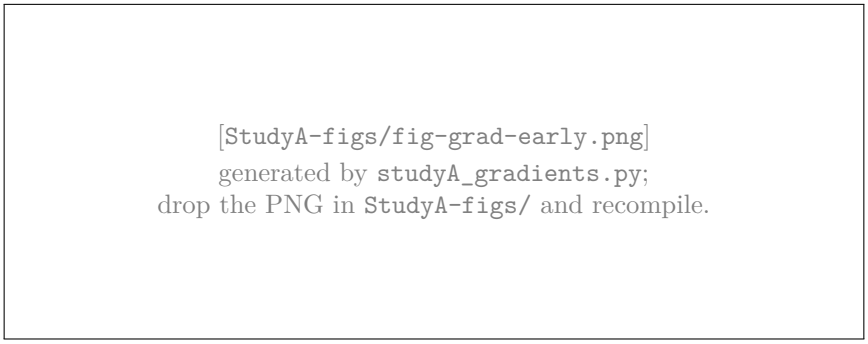


Figure 3: **Early-layer gradient over training.** Gradient norm at the first conv (`inc`) vs. epoch, skip vs. no-skip.

feature path; for a 2D localization heatmap the latter dominates, consistent with the larger tail-error improvement. This directly supports the thesis's choice of a U-Net student and reframes the Mamba failure as a loss of 2D spatial locality rather than a lack of capacity.